Bogeng Song 宋伯耕

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PhD Student, Psychology · Georgia Institute of Technology · Advisor: Dobromir Rahnev

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EDUCATION

PhD in Psychology — Georgia Institute of Technology — Present
Advisor: Dobromir (Doby) Rahnev
Computational cognitive science & perceptual decision-making

MA in Psychology — New York University
Worked with Grace Lindsay, Marisa Carrasco, and Jonathan Winawer

TRAINING & SUMMER SCHOOLS

NeuroAI — Neuromatch Academy — Summer 2025
Deep learning as a framework for understanding brains and behavior

Computational Neuroscience — Neuromatch Academy — Summer 2022
Models of neural dynamics, decision-making, and learning

PUBLICATIONS & PRESENTATIONS

† indicates equal contribution; underlined links are clickable.

Journal Articles

J2. R. Ezzo, B. Song, B. Rokers, & M. Carrasco. Eyes on hold: motion task difficulty jointly delays microsaccade and pupil responses. Scientific Reports. Article · DOI — 2025

J1. X. Lu†, B. Song†, S. Zhang, S. Zhang, M. Huang, Y. Wang, & Y. Jiang. Implied gravity promotes coherent motion perception. npj Microgravity. Article · DOI — 2025

Preprints

P1. N. C. Benson, B. Song, S. Chen, T. Miyata, H. Takemura, & J. Winawer. Machine learning matches human performance at segmenting the human visual cortex. bioRxiv. Preprint · DOI — 2025

Conference Presentations

C4. B. Song, & D. Rahnev. The influence of response bias on confidence and accuracy in multi-alternative tasks for humans and artificial neural networks. Vision Sciences Society (VSS); Journal of Vision. — 2025

C3. Q. Yang, H. W. Han, B. Song, J. Golomb, D. Rahnev, Y. Mohsenzadeh, & H.-H. Li. Hierarchical representational transformations of working memory in brains and machines. Cognitive Computational Neuroscience (CCN). — 2025

C2. N. C. Benson, B. Song, T. Miyata, H. Takemura, & J. Winawer. Automated delineation of visual area boundaries and eccentricities by a CNN using functional, anatomical, and diffusion-weighted MRI data. MODVIS (VSS satellite workshop). — 2023

C1. R. Ezzo, B. Song, B. Rokers, & M. Carrasco. Microsaccade rates reflect trial difficulty for perifoveal motion discrimination. Vision Sciences Society (VSS); Journal of Vision. 2023
Abstract

SELECTED RESEARCH PROJECTS

How bio-inspired attention affects task performance in visual and auditory models — Lindsay Lab, New York University

Added feature-similarity attention to matched visual and auditory networks; attention improved visual but not auditory task performance.

Motion discrimination around the visual field — Carrasco Lab, New York University

Psychophysics and eye-tracking measuring how motion sensitivity varies across the visual field.

COURSEWORK & COURSE PROJECTS

Differential contributions of episodic and semantic memory to story-telling — Computational Cognitive Modeling (Prof. Brenden Lake), NYU

ML classification of recalled / imagined / retold stories (hippoCorpus); feature importance linked to episodic vs. semantic memory theory.

Bayesian semi-supervised learning with function-space variational inference — Bayesian Machine Learning, NYU

Combined function-space variational inference with maximum-uncertainty regularization; evaluated on SVHN and CIFAR-10.

Other relevant coursework

Machine Learning (NYU, Center for Data Science) · Artificial Intelligence (Georgia Tech, School of Computer Science)

SKILLS

- Programming: Python, MATLAB, R
- Machine learning: PyTorch, scikit-learn, deep neural network modeling, Bayesian methods
- Scientific computing: NumPy, SciPy, Pandas, Matplotlib
- Methods: Psychophysics, eye-tracking, computational & cognitive modeling, fMRI analysis
- Tools: PsychoPy, Git/GitHub, LaTeX